FinAgent: Automated SEC 10-K Risk Analysis via Retrieval-Augmented Generation

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Abstract

Analyzing SEC 10-K filings means working across two very different kinds of data, structured financial tables and long unstructured risk narratives, and doing it manually across multiple companies is slow even for experienced analysts. We built **FinAgent**, an end-to-end pipeline that automates peer company identification and comparative risk analysis directly from SEC EDGAR filings. The system ingests structured financial data through SEC XBRL APIs, ranks risk sentences using FinBERT, extracts and maps risk signals to a six-domain taxonomy using an LLM, and retrieves semantically similar companies via FAISS with 1024-dimensional BGE-M3 embeddings. All financial delta computations including ratios, year-over-year changes, and severity labels are handled entirely in code, a deliberate choice to keep those calculations strict, auditable, and reproducible rather than leaving numerical reasoning to a language model. We scoped the dataset to U.S. technology sector companies because industry-level comparison produces the strongest signal, companies operating in the same space share regulatory exposure, revenue structures, and risk language, which makes peer matching meaningful. Within that scope we deliberately included companies with varied risk profiles, from large stable platforms like Apple and Alphabet to companies with more volatile trajectories, to ensure the system was tested against a real spread of financial conditions. FinAgent shows that a deterministic, rule-governed pipeline with targeted LLM assistance can produce financial risk assessments that are grounded and comparable.

1 Introduction

Public companies in the United States are required to file annual reports, known as **Form 10-K**, with the U.S. Securities and Exchange Commission (SEC). These filings provide a comprehensive view of a company's financial performance and risk exposure, combining structured financial statements with extensive unstructured disclosures such as management discussion and risk factors. While 10-K reports are a critical source of information for investors and analysts, their length and mixed structure make systematic analysis difficult and peer comparisons painfully slow even for experienced analysts.

Recent Generative AI systems, particularly retrieval-augmented generation (RAG), have improved the ability to summarize long documents. However, 10-K filings present a structural challenge that flat RAG pipelines are not well equipped to handle. These documents mix numerical financial data,

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boilerplate legal language, and qualitative risk disclosures across hundreds of pages, and a pure similarity fetch has no mechanism to distinguish between them. Without any filtering layer, retrieved chunks are as likely to contain routine business boilerplate as they are to contain meaningful risk signals, and in large documents context dilution makes this worse since the most relevant content gets buried under everything else the retrieval pulls in. For financial analysis specifically, this is a reliability problem because numerical trends and risk signals need to be isolated, weighted, and traced back to their source before any comparison across firms is meaningful.

We present **FinAgent**, a hybrid pipeline for structured SEC 10-K analysis that combines deterministic financial data processing with structured risk extraction and peer company retrieval. Rather than retrieving over raw document chunks, **FinAgent** retrieves over curated risk signals that have already been filtered, labeled, and structured before any retrieval happens. The system ingests annual filing data from SEC EDGAR including XBRL-derived financial metrics and key 10-K sections, scores and filters risk sentences using FinBERT, and applies taxonomy-based curation to transform raw filing evidence into structured risk signals. All financial delta computations are handled entirely in code, keeping numerical reasoning strict and auditable. These structured representations are then embedded using BGE-M3 and indexed in FAISS to enable peer company retrieval based on combined financial and risk similarity.

We validated the system on U.S. technology sector companies because within-industry comparison produces more verifiable peer matches. Companies in the same sector share regulatory exposure, revenue structures, and risk language, which makes similarity meaningful rather than coincidental.

The core contribution of this work is an architectural comparison between a *structured multi-stage pipeline* and a *flat single-step RAG pipeline* for financial report generation. Both pipelines use the same language model and output schema, which lets us examine the impact of pipeline structure rather than model choice. This work evaluates whether structured intermediate representations improve downstream reasoning quality, and whether embedding-based retrieval over curated financial and risk signals yields different peer comparisons than flat retrieval over unprocessed text.

2 Literature Review

Understanding what makes financial document analysis hard was the starting point for how we designed FinAgent. Benchmark datasets like FinQA [3] and TAT-QA [8] make it clear that financial tasks are not simple extraction problems. They require multi-step numerical reasoning and the ability to work across tables and text at the same time. More recent evaluations confirm that LLMs still struggle with arithmetic reasoning and compositional inference in document-based settings [7]. This told us early on that handing raw filing text to a model and asking it to reason across companies was not going to be reliable enough, which pushed us toward separating the numerical computation from the language model entirely.

At the same time, prior work shows LLMs are genuinely useful for certain financial tasks when the input is well structured. Kim et al. [5] demonstrate that generative models can produce useful narrative insights from structured financial data and hold up competitively on prediction tasks. Most of this work focuses on single company tasks like prediction or sentiment classification. Structured comparative analysis across multiple firms is largely absent, and that is the gap FinAgent targets.

On the representation and retrieval side, FinBERT [1] showed that domain adaptation matters for financial text. A generic embedding model treats risk disclosures the same as any other text, but a model trained on financial language picks up on the specific signals that matter in that domain. That is why we used FinBERT for sentence scoring rather than a general purpose model, we needed something that could distinguish meaningful risk language from boilerplate before anything gets passed downstream. For retrieval infrastructure, BGE-M3 [2] gave us high quality 1024-dimensional embeddings and FAISS [4] gave us the similarity search layer on top. Existing work typically applies these components over entire documents. We applied them after isolating risk-relevant sections from the filing, which reduced processing overhead and ensured retrieval was working over targeted content rather than the full document.

Most finance-focused evaluation measures prediction accuracy or sentiment classification, neither of which reflects what an analyst actually needs. FinAgent is not designed to replace analyst judgment but to support it. By generating citation-grounded comparison reports [6], the system gives analysts a

structured starting point that covers risk overlap and financial trajectory across peers, reducing the research burden and letting them focus on the parts that actually require their expertise.

3 Methodology

3.1 System Overview and Research Design

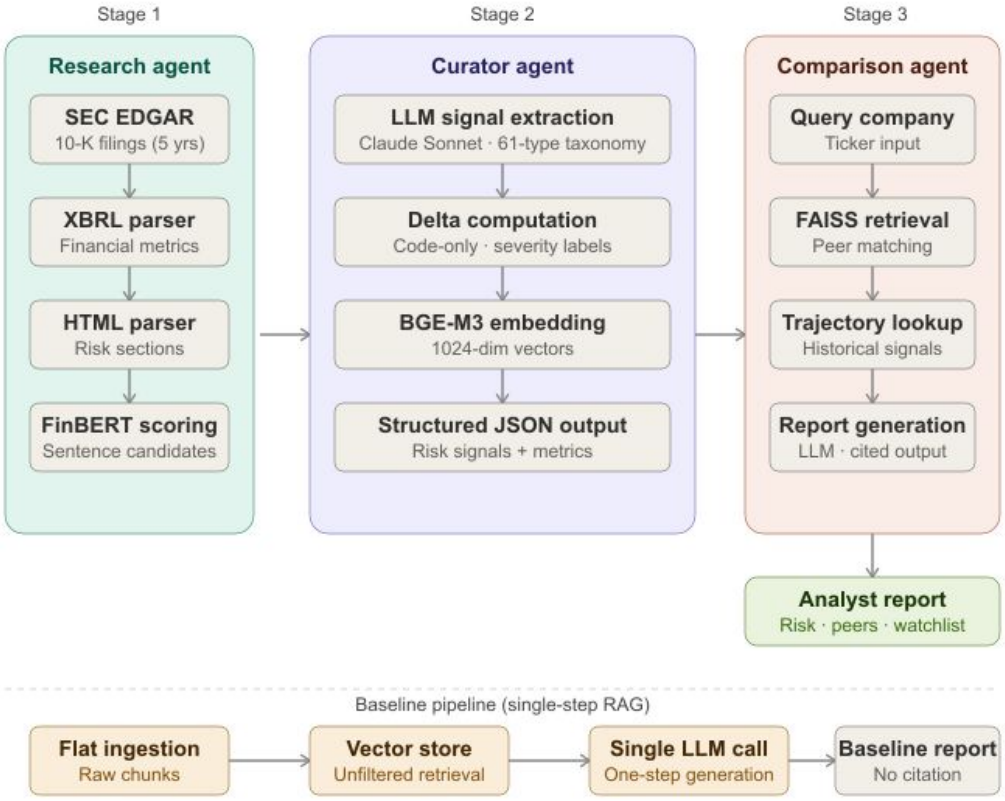

Figure 1: FinAgent system architecture. The pipeline separates structured ingestion, risk curation, and peer retrieval across three specialized agents, compared against a flat single-step RAG baseline.

FinAgent is designed as a comparative system for SEC 10-K analysis, built around a controlled comparison between two report generation strategies: a *structured multi-stage pipeline* and a *flat single-step RAG baseline*, as shown in Figure 1.

We initially considered an explicit tool-calling agent architecture for the upstream data preparation workflow but kept it fully deterministic instead. The procedure is linear, debugging was straightforward, and intermediate artifacts stayed directly inspectable against source filings.

The structured pipeline passes data through ingestion, FinBERT-based sentence scoring, LLM-assisted signal extraction, code-driven delta computation, and BGE-M3 embedding before retrieval and report generation. The baseline skips these intermediate stages entirely, retrieving directly over raw ingestion artifacts and generating the report in a single LLM call.

Both pipelines use the same language model and output schema so that differences in report quality are attributable to pipeline structure rather than model choice. Financial filings mix structured numerical data with boilerplate legal language and qualitative risk disclosures, and a flat retrieval

approach has no way to separate them, which is exactly what makes pipeline design the primary object of study in this work.

3.2 Data Collection and Deterministic Ingestion

The first stage gathers financial and filing data directly from SEC sources without any LLM involvement. The output is a `complete_ingestion.json` file generated per ticker, containing company identity, annual financial metrics, filing metadata, and selected 10-K text sections. This artifact serves as the source of truth for everything downstream.

Structured financial values are derived from SEC Company Facts and XBRL-aligned disclosures. The pipeline standardizes annual metrics including revenue, net income, cash, assets, long-term debt, operating cash flow, R&D expense, and gross profit where available. Because SEC disclosures vary in naming conventions and reporting formats across companies, the ingestion process includes data cleaning, fallback tag handling, and deduplication to produce a single canonical datapoint per fiscal year. This normalization is what makes cross-company comparisons consistent later in the pipeline.

In parallel, the ingestion stage extracts key 10-K narrative sections relevant to risk and performance analysis, including Item 1A (Risk Factors), Item 3 (Legal Proceedings), Item 7 (Management Discussion and Analysis), Item 7A (Market Risk), and Item 1C (Cybersecurity) where available. Section boundaries are identified using rule-based heading detection and stop-boundary logic rather than generative extraction, which keeps the data acquisition step fully reproducible and avoids introducing hallucination risk before the pipeline has even started.

3.3 Risk Extraction and Curator Representation

After deterministic ingestion, the second stage transforms each filing into a structured yearly extraction artifact. For each company and filing year, the extraction pipeline processes selected 10-K sections, splits them into sentences, filters out low-value content such as headings, boilerplate fragments, and table-like text, and scores the remaining sentences using the ProsusAI FinBERT model. FinBERT is used here as a sentence ranker rather than a classifier. The risk score assigned to each sentence corresponds to the model's negative-class probability, which approximates how strongly a sentence signals financial risk. The pipeline retains the highest-ranked sentences along with their surrounding context, preserving the previous and next sentence as evidence spans for downstream stages.

In parallel, the extraction stage computes year-over-year numeric deltas for key financial metrics including revenue, net income, cash, assets, long-term debt, operating cash flow, RD expense, and gross profit where available. Each metric is aligned across the current and prior fiscal year and the percentage change is mapped into discrete severity categories: *strong_growth*, *moderate_growth*, *stable*, *moderate_decline*, and *severe_decline*. This bucketing standardizes financial trends across companies at very different scales and gives the downstream model structured comparable signals rather than raw values it would have to interpret itself.

The curator stage converts the extraction artifact into a retrieval-ready representation. One thing we noticed early on was that ranking purely on FinBERT sentiment scores over-selected generic risk-factor boilerplate from Item 1A while missing more specific and quantified disclosures in other sections. To address this, the curator applies section-aware candidate selection, distributing sentences across analytically relevant sections with greater weight on Item 7 (Management Discussion and Analysis) and Item 7A (Market Risk), followed by legal and cybersecurity sections, and reduced emphasis on Item 1A.

Selected candidates are then passed to Claude Sonnet via a structured curator prompt that extracts risk signals under a controlled 61-type taxonomy. Each signal includes fields for topic, signal type, section, summary, severity, and citation, and outputs are validated against strict Pydantic schemas. Risk signal severity follows a four-level scale: *critical*, *high*, *medium*, and *low*, assigned based on the nature and magnitude of the identified signal. The final curator artifact contains filing metadata, financial delta labels, curated risk signals, and a synthesized text representation used for embedding and retrieval in the next stage.

3.4 Semantic Peer Retrieval

The retrieval stage identifies peer companies based on shared financial and risk patterns captured in the curated filing representation. For each curator artifact, the system constructs an `embedding_text` field by concatenating the company identifier, filing year, discrete financial delta labels, and a capped set of curated risk-signal summaries.

The combined text is embedded using the `BAAI/bge-m3` model, which produces a 1024-dimensional dense vector. These vectors are normalized and stored within the curator artifacts, enabling reuse across retrieval runs. All vectors are indexed using FAISS with an inner-product index, which corresponds to cosine similarity when embeddings are normalized. During retrieval, the system queries the index using the target company-year vector, excludes the query company, and deduplicates multiple filings from the same peer by retaining only the highest-scoring instance. The top distinct companies are selected as peers, giving the report generation stage a grounded set of comparators drawn from actual financial and risk similarity.

3.5 Agentic Report Generation

Once the target company's curator artifact and matched peer curator artifacts are available, the orchestration layer assembles the final comparison context and invokes the report generation step using GPT-5 mini. The most recent filing year of the target company serves as the anchor for comparison, and matched peers contribute not only the aligned filing year but also up to two later curator years where available, giving the report generation stage multi-year trajectory context rather than a single-year snapshot.

The report is written into a structured artifact containing fields such as `summary`, `posture_assessment`, `target_profile`, `peer_snapshot`, `risk_overlap_rows`, `forward_watchlist`, and `narrative_sections`. The model does not reason over raw filing text or flattened financial tables at this stage. It operates entirely over the curated intermediate representations produced earlier in the pipeline, which makes intermediate outputs directly inspectable and traceable back to specific filing evidence rather than relying on the model to interpret raw documents directly.

3.6 Baseline RAG Flow

The baseline pipeline starts directly from the ingestion artifact. It treats the `complete_ingestion.json` outputs as the source of truth, converts the ingestion content into a compact textual representation, and uses ingestion-derived embeddings to retrieve peer companies. To keep the comparison focused on pipeline structure rather than retrieval infrastructure, the baseline uses the same embedding model and FAISS-based similarity search as the agentic system, but applies them to ingestion-level representations instead of curator artifacts. It does not use yearly extraction artifacts, curated risk signals, financial delta labels, or multi-year curator context.

Both pipelines operate over the same targeted 10-K sections rather than the full document, ensuring that observed differences in output quality are attributable to pipeline structure and intermediate representation rather than differences in input scope.

After retrieval, the baseline constructs a prompt from the target company's flattened ingestion representation along with similarly flattened peer context and makes a single language model call to generate the final report. The output follows the same structured schema as the agentic system, including fields such as `summary`, `posture`, `target_profile`, `peer_snapshot`, `risk_overlap_rows`, `forward_watchlist`, and `narrative_sections`. Keeping the schema identical ensures the comparison reflects differences in pipeline structure rather than output format.

3.7 Evaluation Framework

Evaluating long-form financial reports requires assessing both structural correctness and evidence grounding across two very different pipeline architectures. To handle this, we convert each report into a shared `EvaluationInput` representation that extracts the summary, posture label and rationale, target risks, overlap rows, watchlist items, and narrative sections, and constructs deterministic fact

lists and evidence pools. Both pipelines are evaluated under the same scoring interface so that differences in scores reflect pipeline behavior rather than evaluation setup.

Evidence pools differ by pipeline due to architectural differences. For the agentic pipeline, the evaluator reads curator artifacts and derives deterministic facts from financial delta labels, with evidence drawn from curated risk signals and summaries. For the baseline pipeline, the evaluator reads ingestion artifacts, derives financial deltas from the most recent year, discretizes them into labels such as `strong_growth` and `moderate_decline`, and constructs evidence from flattened ingestion text. Peer evidence is added from matched peer artifacts for both pipelines.

3.7.1 Deterministic Scorecard

The deterministic evaluation path does not use a language model. It scores each report across three components: `deterministic_consistency`, `evidence_coverage`, and `overreach_penalty`.

Deterministic consistency is computed as the fraction of rule-based checks satisfied, including structural completeness, plausibility of posture given financial and risk signals, coherence of overlap rows, and presence of supporting evidence in watchlist entries. Evidence coverage is the mean of three ratios: the fraction of narrative sections containing citations, the fraction of target risks with either a citation or summary, and the fraction of watchlist items with peer evidence. If no watchlist items are present, the third term defaults to 1.0. This metric measures how explicitly the report surfaces grounding signals.

Overreach is modeled as a penalty. Fixed penalties are added when watchlist items lack peer evidence, when forecast-like terms such as *will* or *expect* appear without support, or when shared peer patterns are referenced without corresponding overlap rows. The total penalty is clipped to 0.5. These heuristics approximate unsupported extrapolation in a reproducible way without requiring a model call.

3.7.2 LLM Judge Evaluation

When judge-based evaluation is enabled, GPT-5 mini performs two additional semantic evaluation steps on top of the deterministic scorecard.

First, atomic claims are extracted from the summary, posture, and narrative sections. Each claim is evaluated against the evidence pool and assigned a support label mapped to a numeric score as follows:
supported maps to 1.0,
partially supported maps to 0.5,
and *unsupported* maps to 0.0.

The average across all claims defines the `claim_support` score. Second, GPT-5 mini assigns a `comparative_usefulness` label mapped to a score as follows:
strong maps to 1.0,
partial maps to 0.5,
and *weak* maps to 0.0.
Responses are cached using artifact and prompt hashes to ensure consistency across runs.

3.7.3 Final Scoring

The final score is a weighted combination of all active components. The component weights are as follows:
`deterministic_consistency` contributes 0.25,
`evidence_coverage` contributes 0.20,
`claim_support` contributes 0.30 in judge-enabled mode,
and `comparative_usefulness` contributes 0.15 in judge-enabled mode.

The weighted sum is normalized by the active weights and the `overreach_penalty` is subtracted. These weights prioritize claim-level correctness while retaining contributions from structural and coverage metrics.

Evaluation is performed in a ticker-matched setting where the agentic and baseline reports for the same company are compared directly, enabling per-company comparison across all evaluation dimensions.

4 Experimental Results & Discussion

4.1 UI-Based Report Comparison

The system includes a local Streamlit interface that presents side-by-side outputs from the agentic and baseline pipelines for the same target company. The interface exposes multiple comparison views covering report structure, risk signals, peer retrieval, and narrative output.

The top section displays a pipeline quality comparison using a radar chart, along with a bar chart showing risk severity distribution.

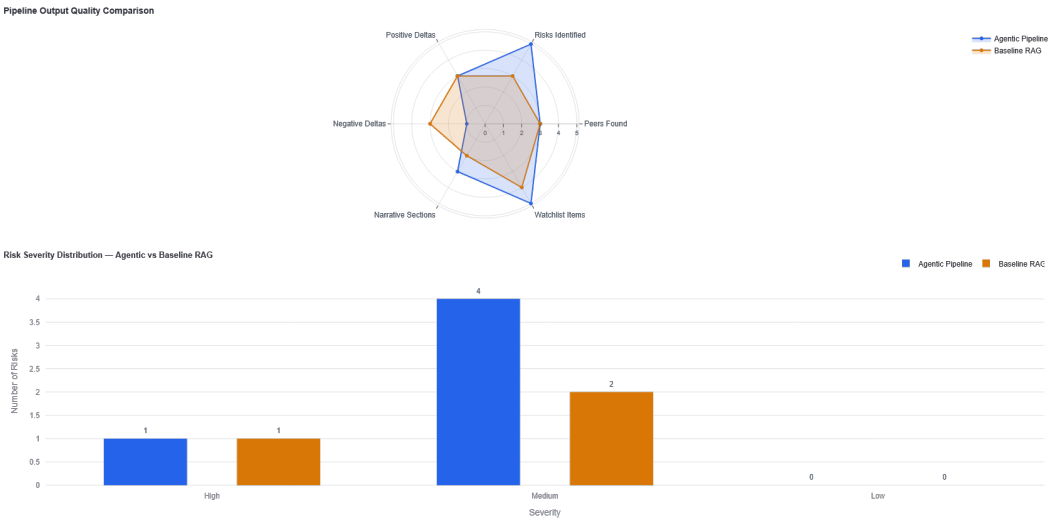

Figure 2: Pipeline quality comparison (up) and risk severity distribution (down).

A tabular view labeled *Risk Identification Diff* lists risk types identified by each pipeline, indicating presence or absence per pipeline.

Risk Identification Diff

Risk Type	Agentic	Baseline RAG
Leverage Risk	–	✓
Liquidity Risk	–	✓
Profitability Risk	–	✓
Active Litigation	✓	–
Content Moderation Liability	✓	–
Foreign Currency Exposure	✓	–
Intensifying Competition	✓	–
Interest Rate Sensitivity	✓	–

Figure 3: Risk identification differences between pipelines.

The *Risk Overlap by Group* table summarizes shared, target-only, and peer-only risk categories.

The interface also includes a *Peer Match Comparison* table, which lists retrieved peer companies along with similarity scores for each pipeline.

A *Posture & Summary* section presents the generated posture classification and summary text for both pipelines.

Additional sections in the interface include financial metric visualizations, positive and negative deltas, top risks with evidence, peer similarity views, risk overlap summaries, forward watchlist items, and narrative report sections.

Figure 2 through Figure 5 illustrate representative views from the interface. A complete comparison report for Snapchat is available in the documentation folder.

Risk Overlap by Group

Group	Agentic	Baseline RAG
Shared Now	Active Litigation, Foreign Currency Exposure, Intensifying Competition, Interest Rate Sensitivity, Negative Operating C	Increasing R&D Spend, Long Term Debt Increases
Target Only Now	Content Moderation Liability, Pricing Power Erosion, Regulatory Investigation	Negative Net Income, Low Cash Liquidity
Peer Only Now	Business Model Transition Risk, Customer Churn Risk, Cybersecurity Vulnerability, Infrastructure Single Point Failure, I	Very High Absolute Profitability, Very High Operating Cash Flow

Peer Match Comparison

Peer Ticker	Agentic Similarity	Baseline Similarity
AAPL	—	0.8202
ADBE	0.7476	0.7983
CRM	0.7588	—
META	—	0.8236
UBER	0.7395	—

Figure 4: Risk overlap grouped by category.

Posture & Summary

Agentic Pipeline

Posture: Stable / Strong

- Analysis identifies 1 high-severity risk (interest_rate_sensitivity) and 1 negative financial delta (NetIncome: -34.01).
- Three positive deltas (OperatingCashFlow strong_growth +58.69; Revenues moderate_growth +10.63; R&D moderate_growth +6.02) provide a financial buffer.
- Six risk types are shared with peers, indicating broad industry exposure rather than firm-specific concentration.

Warnings

Summary

Snap Inc. presents a stable / strong posture versus its matched peer neighborhood, supported by strong operating cash flow and revenue growth but offset by a severe decline in net income. The company has three positive financial deltas (OperatingCashFlow: +58.69, Revenues: +10.63, ResearchAndDevelopment: +6.02) and one negative delta (NetIncome: -34.01). A high-severity interest-rate sensitivity risk (interest expense rise to $122M in 2025) and several medium-severity operational and litigation risks merit continued attention.

Step Status

Baseline RAG

Posture: Mixed

- Revenue expanded by $1,814M from FY2021 to FY2025 and FY2025 revenue increased $570M YoY, indicating continued top-line momentum.
- Profitability has improved materially from the FY2022 trough (net loss $-1,430M) to FY2025 ($-460M), and operating cash flow strengthened to $656M in FY2025.
- Liquidity and leverage remain areas of concern: cash fell to $1,030M in FY2025 and long-term debt is sizable at $3,537M (debt level roughly 46% of FY2025 assets $7,678M).

Summary

Snap shows steady revenue growth (FY2021 $4,117M → FY2025 $5,931M) and a materially improved operating cash flow (FY2025 $656M), but remains unprofitable (net loss $-460M in FY2025) with limited liquidity (cash $1,030M) and a relatively high long-term debt stock ($3,537M) versus its asset base.

Step Status

Figure 5: Posture and summary comparison.

4.2 Evaluation Results

We evaluated five ticker-matched report pairs using the shared evaluation framework with judge-based scoring enabled. The evaluated companies were `AAPL`, `AMD`, `BYND`, `LYFT`, and `SNAP`. Tables 1, 2, and 3 summarize the results.

Under deterministic evaluation (Table 1), the agentic pipeline achieves near-perfect overall scores across all tickers, with values ranging from 0.9074 to 1.0000. In contrast, the baseline pipeline remains consistent at 0.8519 across all companies. The agentic pipeline also achieves perfect evidence coverage (1.0000) across all cases, while the baseline maintains a fixed coverage score of 0.6667.

Under judge-enabled evaluation (Table 2), the baseline pipeline achieves higher overall scores across all five tickers. Baseline overall scores range from 0.5426 to 0.6759, while the agentic pipeline ranges from 0.2389 to 0.5139. The largest gap appears for `AAPL`, where the baseline reaches 0.6759 compared to 0.2389 for the agentic pipeline.

At the metric level, both pipelines maintain high deterministic consistency, with the baseline achieving 1.0000 across all cases and the agentic pipeline showing a single deviation (0.8333 for `LYFT`). Evidence coverage remains consistently higher for the agentic pipeline (1.0000) compared to the baseline (0.6667). In contrast, the baseline pipeline achieves higher claim support and usefulness scores, with claim support reaching up to 1.0000 and averaging 0.9400, compared to a maximum of 0.5833 and an average of 0.4583 for the agentic pipeline. Overreach penalties are lower for the baseline, ranging from 0.2000 to 0.3000, while the agentic pipeline ranges from 0.2500 to 0.4000.

Average scores across all tickers (Table 3) show that the agentic pipeline performs better under deterministic evaluation (0.9815 vs. 0.8519), while the baseline pipeline achieves higher overall scores under judge-enabled evaluation (0.6226 vs. 0.4069).

4.3 Observations and Interpretation

The evaluation highlights the difficulty of assessing long-form financial reports, where both structural correctness and evidence-grounded reasoning must be considered. While deterministic metrics provide stable signals, the judge-enabled evaluation introduces variability due to its reliance on LLM-based claim extraction and semantic matching.

Table 1: Deterministic evaluation results for five ticker-matched agentic and baseline report pairs. Higher is better except for overreach penalty, where lower is better.

Ticker	Pipeline	Overall	Consistency	Coverage	Overreach
AAPL	Agentic	1.0000	1.0000	1.0000	0.0000
AAPL	Baseline	0.8519	1.0000	0.6667	0.0000
AMD	Agentic	1.0000	1.0000	1.0000	0.0000
AMD	Baseline	0.8519	1.0000	0.6667	0.0000
BYND	Agentic	1.0000	1.0000	1.0000	0.0000
BYND	Baseline	0.8519	1.0000	0.6667	0.0000
LYFT	Agentic	0.9074	0.8333	1.0000	0.0000
LYFT	Baseline	0.8519	1.0000	0.6667	0.0000
SNAP	Agentic	1.0000	1.0000	1.0000	0.0000
SNAP	Baseline	0.8519	1.0000	0.6667	0.0000

Table 2: Judge-enabled evaluation results for five ticker-matched agentic and baseline report pairs. Higher is better except for overreach penalty, where lower is better.

Ticker	Pipeline	Overall	Consistency	Coverage	Claim Support	Usefulness	Overreach
AAPL	Agentic	0.2389	1.0000	1.0000	0.1667	0.5000	0.4000
AAPL	Baseline	0.6759	1.0000	0.6667	1.0000	1.0000	0.2500
AMD	Agentic	0.5139	1.0000	1.0000	0.5417	0.5000	0.2500
AMD	Baseline	0.6093	1.0000	0.6667	0.8000	1.0000	0.2500
BYND	Agentic	0.4000	1.0000	1.0000	0.5000	0.5000	0.3500
BYND	Baseline	0.6426	1.0000	0.6667	1.0000	0.5000	0.2000
LYFT	Agentic	0.4037	0.8333	1.0000	0.5000	0.5000	0.3000
LYFT	Baseline	0.6426	1.0000	0.6667	0.9000	1.0000	0.2500
SNAP	Agentic	0.4778	1.0000	1.0000	0.5833	0.5000	0.3000
SNAP	Baseline	0.5426	1.0000	0.6667	1.0000	0.5000	0.3000

Across results, the agentic pipeline consistently achieves higher `evidence_coverage`, while the baseline pipeline achieves higher `claim_support` and `comparative_usefulness`. However, these differences should be interpreted with caution, as judge-based metrics depend on how claims are extracted and matched to evidence, which may favor simpler and more direct outputs.

Qualitative inspection through the UI provides an additional perspective on these differences. The agentic pipeline produces reports with more structured intermediate signals, including clearer risk grouping, explicit overlap categories, and more organized peer context. These characteristics reflect the staged preprocessing and curation steps in the pipeline, which progressively clean and structure the input before generation.

From a systems perspective, these design choices also address practical constraints. Embedding full 10-K filings or operating directly on raw document text is computationally expensive and introduces significant noise. Similarly, passing large, unfiltered contexts into the LLM increases latency, cost, and the likelihood of hallucination. The agentic pipeline mitigates these issues by reducing the input to curated, task-relevant representations before retrieval and generation.

At the same time, the higher `overreach_penalty` observed for the agentic pipeline indicates that richer structure does not guarantee stronger final claims under the current evaluation. Reports with more detailed or forward-looking content are more likely to trigger penalties when explicit support is not detected.

Overall, these results suggest that LLM-based evaluation may not fully capture the benefits of structured intermediate processing. While the agentic pipeline improves the organization and exposure of signals, the evaluation framework places greater emphasis on direct claim–evidence alignment. In practice, assessing the usefulness of such reports may require human evaluation against real business or analytical requirements, where structured context and interpretability play a larger role.

Table 3: Average scores across five tickers for deterministic-only and judge-enabled evaluation. Higher is better except for overreach penalty, where lower is better.

Evaluation	Pipeline	Overall	Consistency	Coverage	Claim Support	Usefulness	Overreach
Deterministic	Agentic	0.9815	0.9667	1.0000	–	–	0.0000
Deterministic	Baseline	0.8519	1.0000	0.6667	–	–	0.0000
Judge-enabled	Agentic	0.4069	0.9667	1.0000	0.4583	0.5000	0.3200
Judge-enabled	Baseline	0.6226	1.0000	0.6667	0.9400	0.8000	0.2500

5 Conclusion and Future Work

5.1 Conclusion

In this work, we developed FinAgent, a modular agentic pipeline for automated analysis of SEC 10-K filings, designed to support structured financial comparison across companies. The system separates ingestion, extraction, curation, retrieval, and report generation into distinct stages, allowing each component to be adapted to different business requirements. We compared this structured pipeline against a baseline retrieval-augmented generation approach under a controlled evaluation framework. The results show that while the agentic pipeline improves evidence organization and structured signal exposure, it does not consistently outperform the baseline in judge-based evaluation metrics. This highlights both the complexity of financial report generation and the limitations of current evaluation methods, particularly when relying on LLM-based judgment for assessing analytical quality.

5.2 Future Work

Future work will focus on improving both the system design and the evaluation methodology. On the system side, we aim to refine extraction and curation stages to better align with specific business requirements, enabling more targeted and relevant signal selection. On the evaluation side, developing more reliable comparison frameworks, including human-in-the-loop assessment, will be critical for accurately measuring report quality. Additionally, the current deterministic pipeline can be extended with tool-based orchestration or adaptive agent workflows, allowing more flexible execution and dynamic decision-making across stages. These directions aim to improve both the reliability and practical applicability of structured financial report generation systems.

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